

ENV 6128

Smart Air Quality Monitoring & Air Pollution Control

Today's Topics

- Number measurement technologies for ambient aerosol
- Expectation
 - Understand the working principles of SMPS and aerodynamic sizers
 - Understand the working principles of PILS & AMS

SMPS

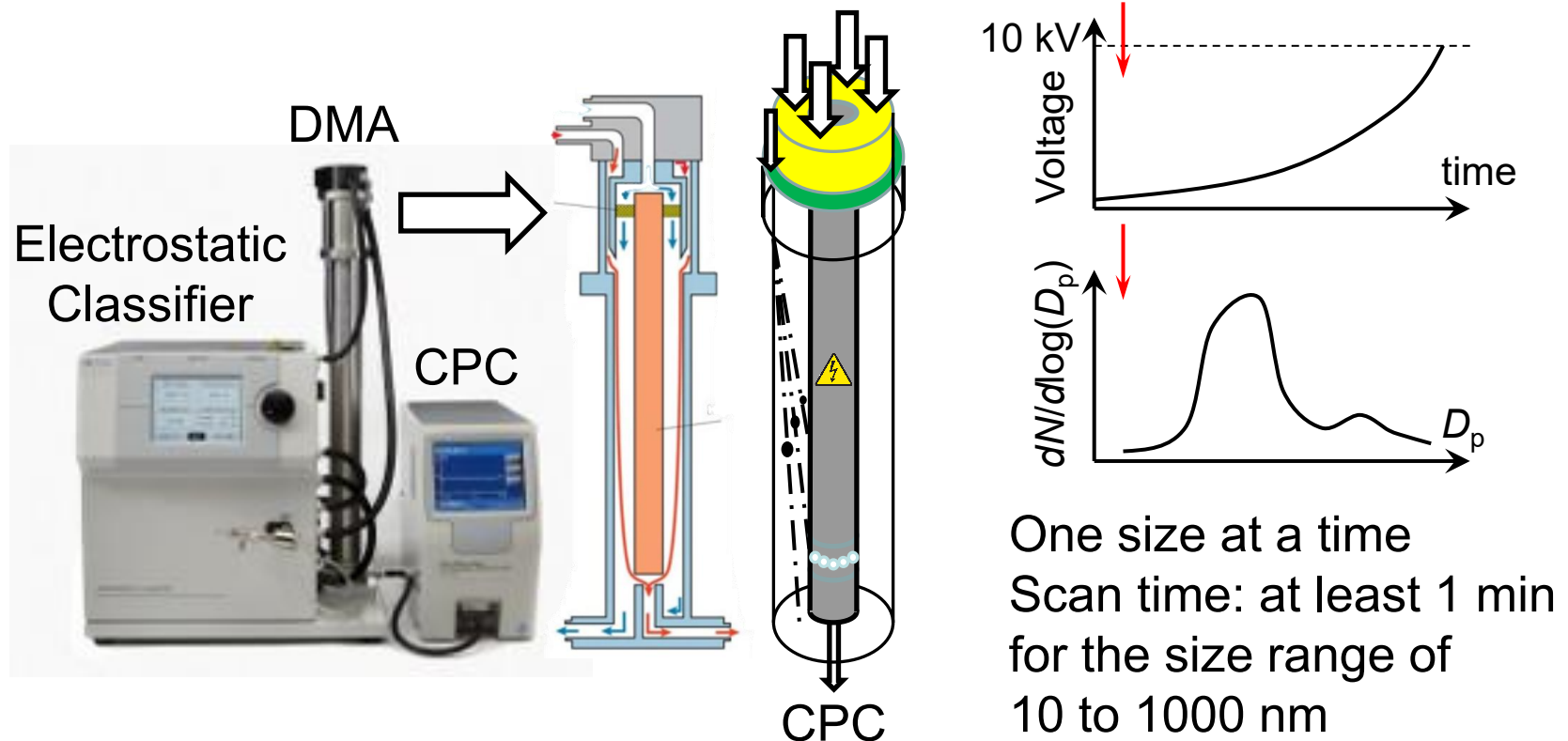
Scanning mobility particle sizers (SMPS)

Submicron aerosol size distribution measurements ($D_p < 1000$ nm)



SMPS

- Scanning mobility particle sizer (SMPS)
- Most used instrument for particle size measurement



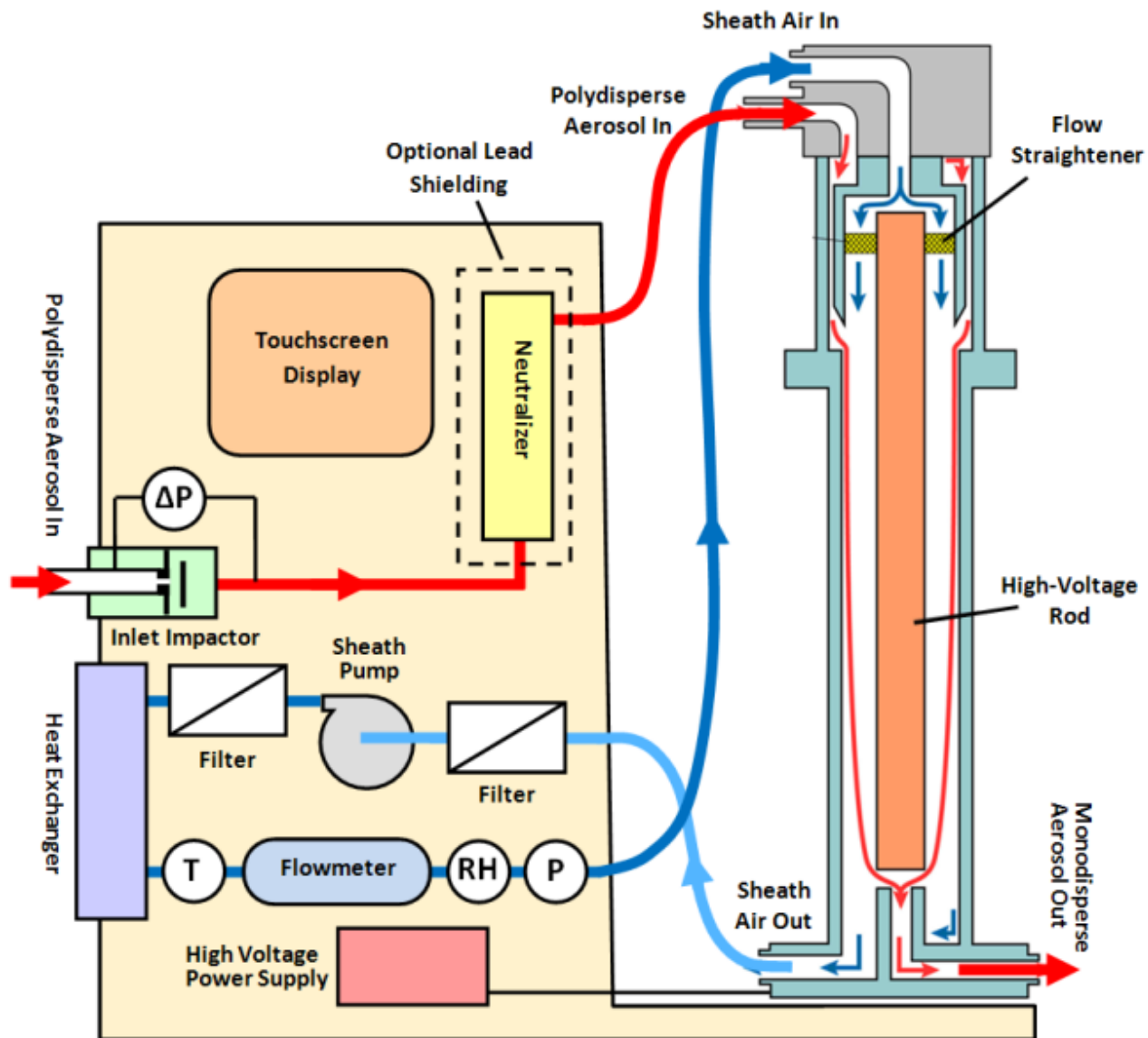


Figure B-3
Flow Schematic for the Electrostatic Classifier with Long DMA

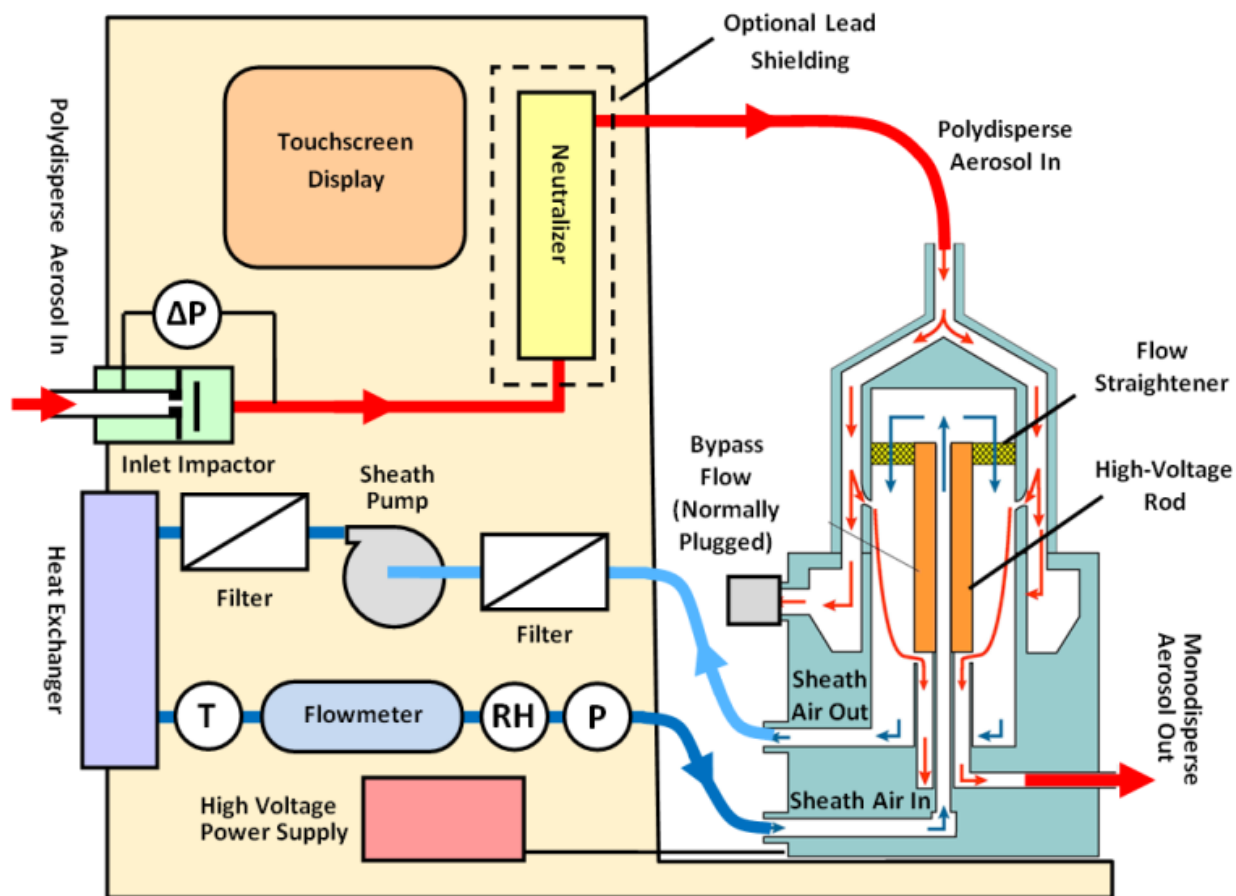
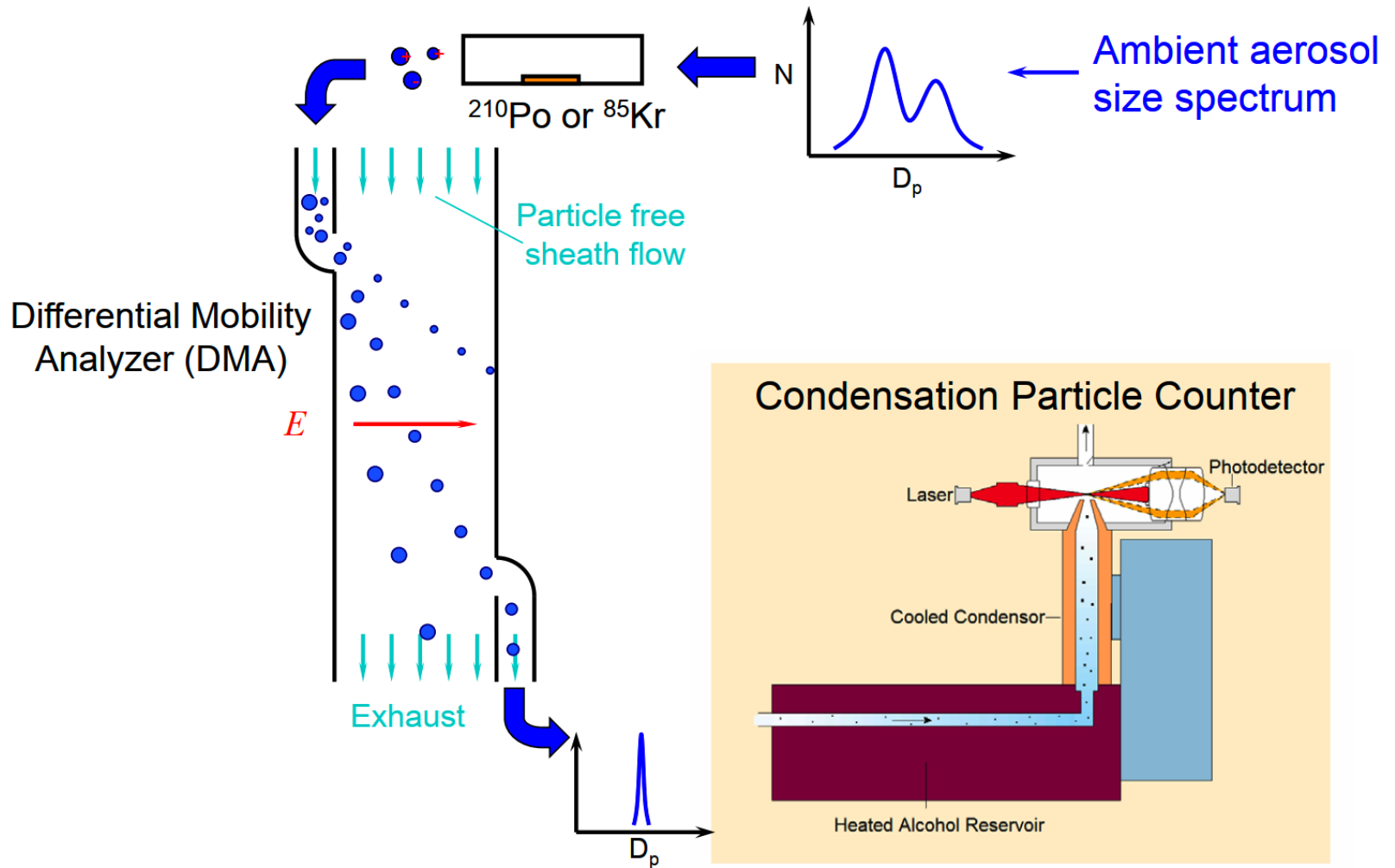
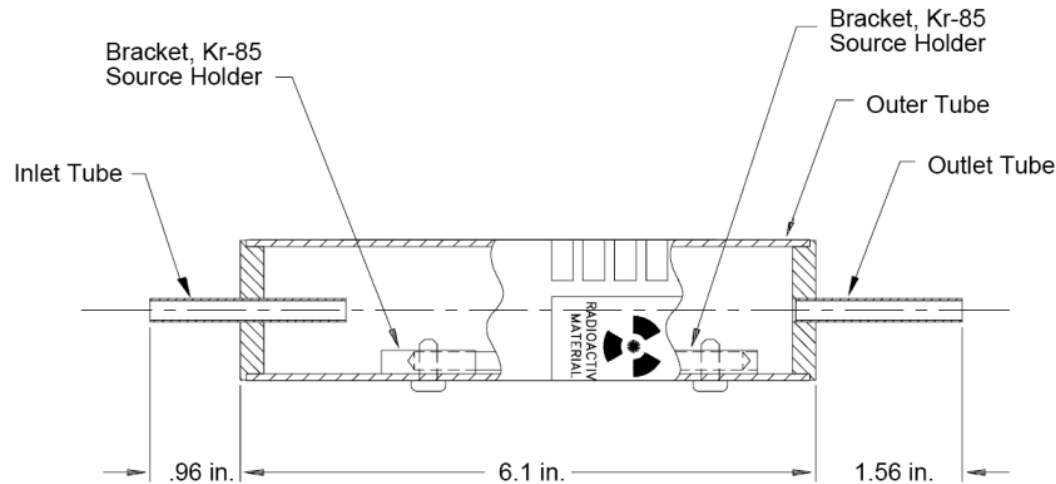


Figure B-4
Flow Schematic for the Electrostatic Classifier with Nano or 1nm-DMA

Scanning mobility particle sizers (SMPS)

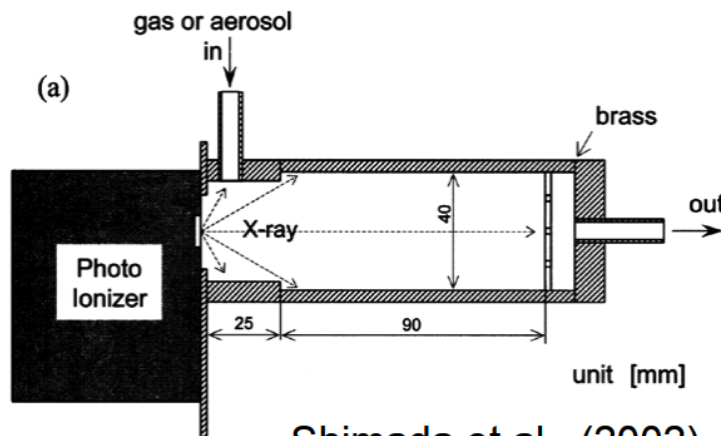


- Kr^{85} (Radiation type: β and γ ; Half life: 10.76 years) and Po^{210} (Radiation type: α ; Half life: 138.4 days)



TSI Kr85 aerosol neutralizer.

- Soft X-ray neutralizer



Shimada et al., (2002)



Unipolar diffusion charging

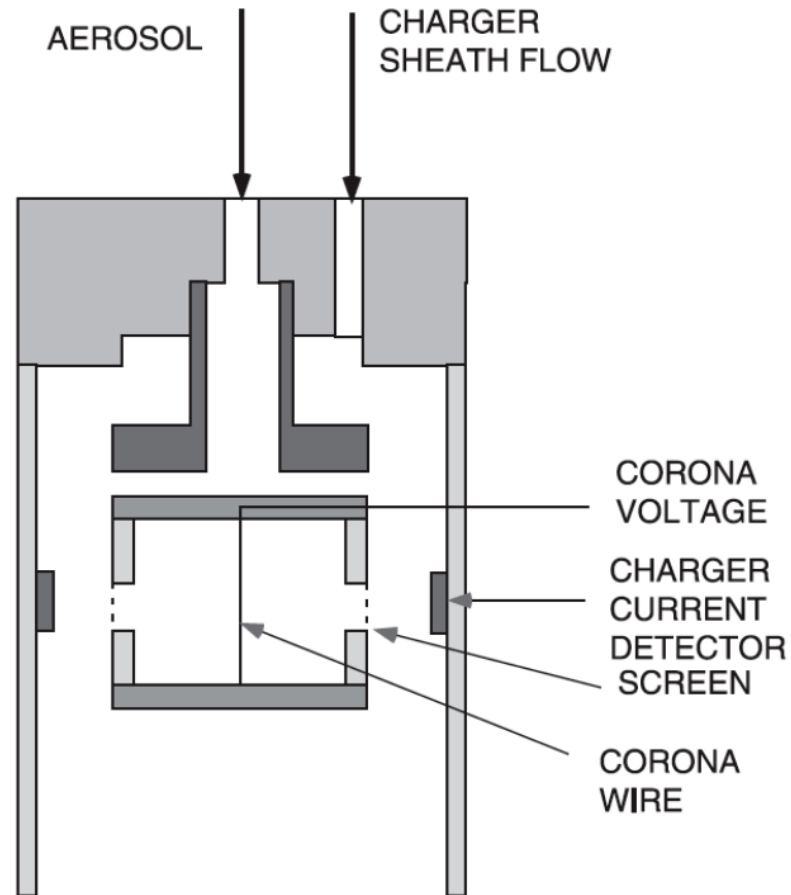
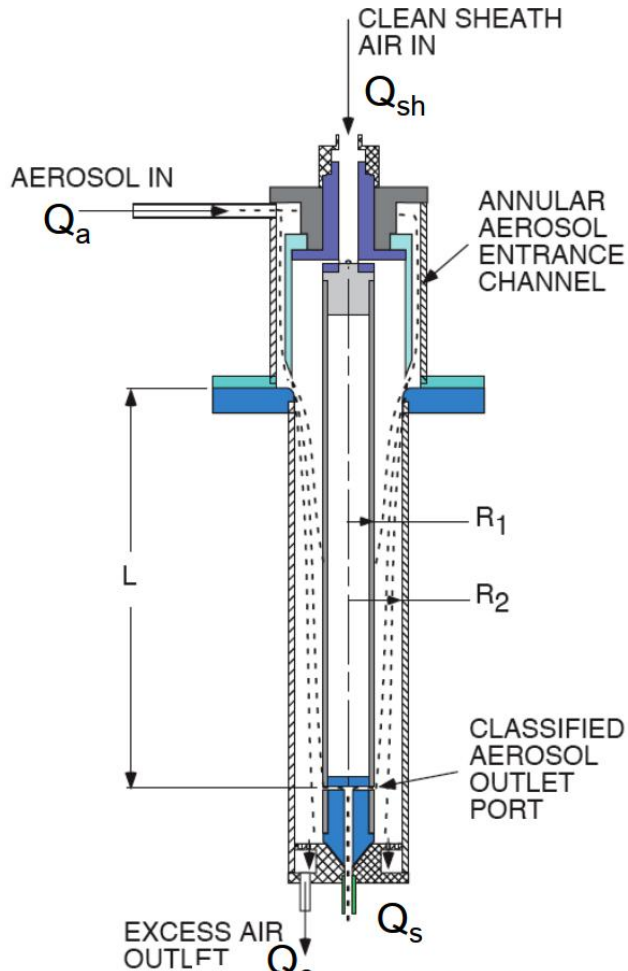


Figure 15-7 Unipolar diffusion charger used in the TSI electrical aerosol analyzer. Reprinted from Flagan (2001) with permission.

Differential mobility analyzer (DMA)



Cylindrical DMA

TSI long
DMA



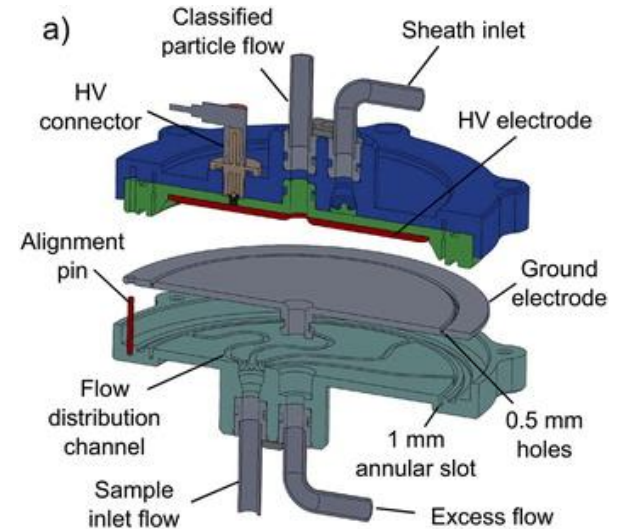
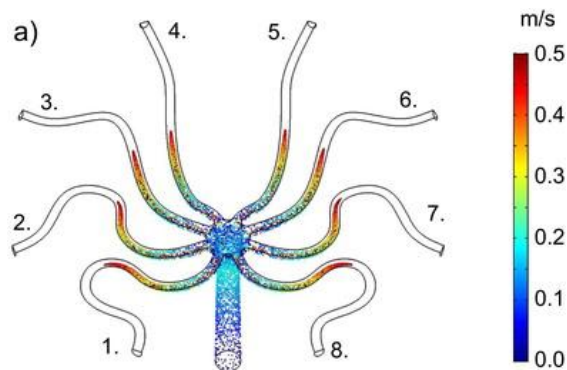
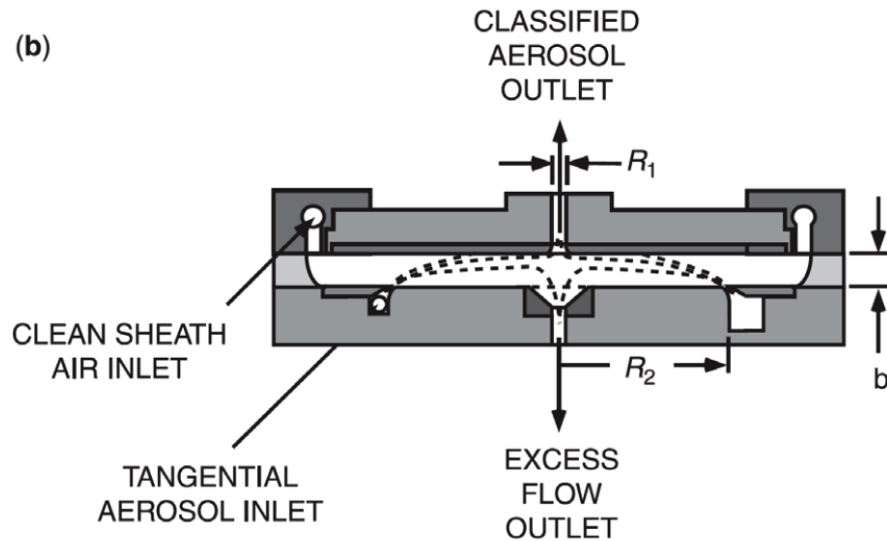
$R_2 = 1.961\text{cm}$
 $R_1 = 0.937\text{cm}$
 $L = 44.369\text{cm}$

TSI Nano
DMA



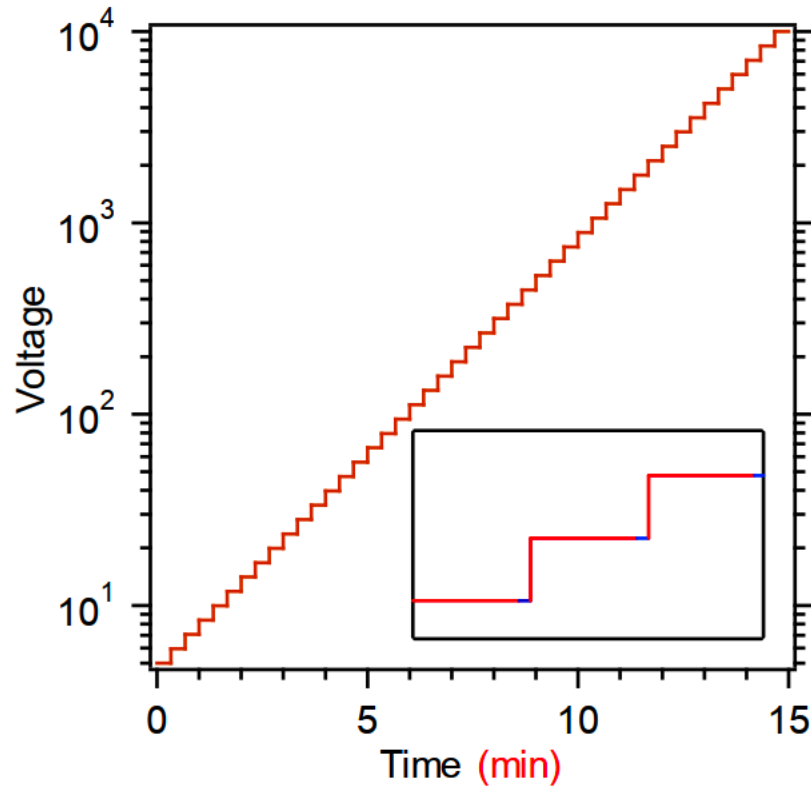
$R_2 = 1.905\text{cm}$
 $R_1 = 0.937\text{cm}$
 $L = 4.987\text{cm}$

Portable/Miniaturized DMA

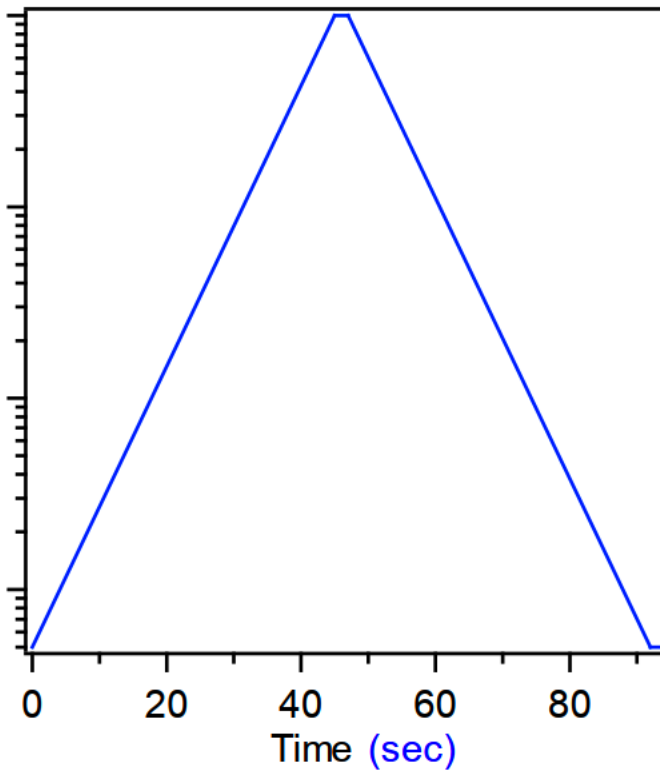


Operation modes

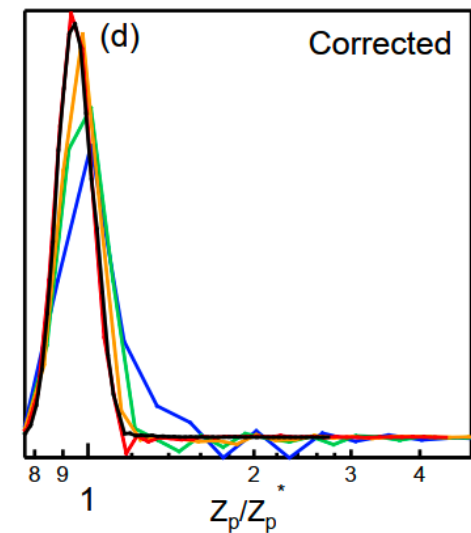
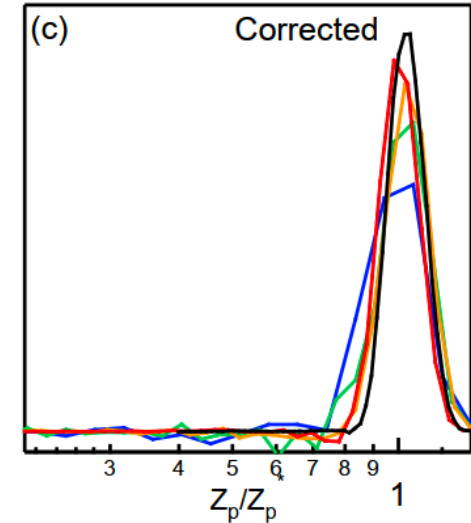
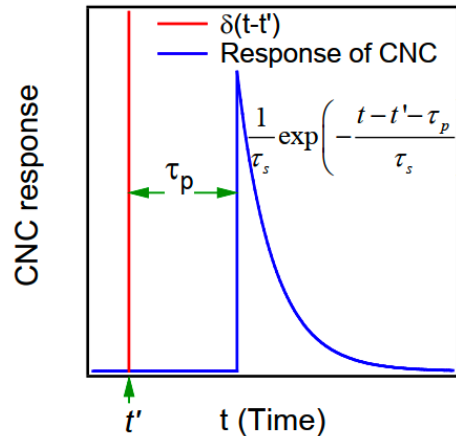
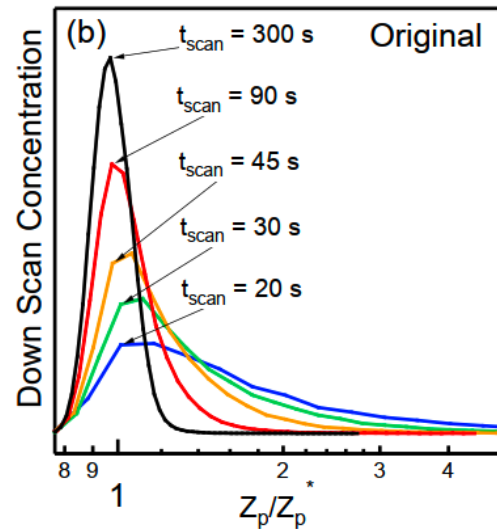
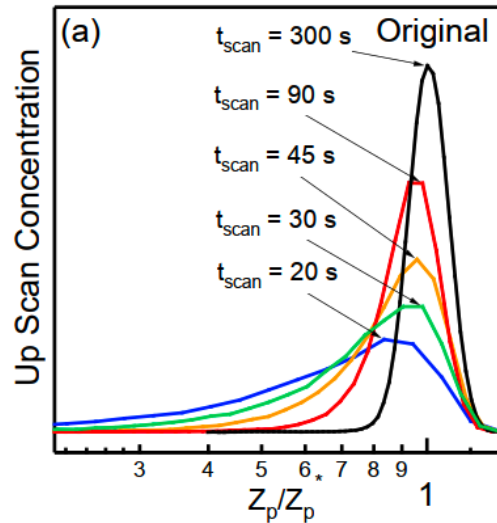
Stepping mode



Scanning mode

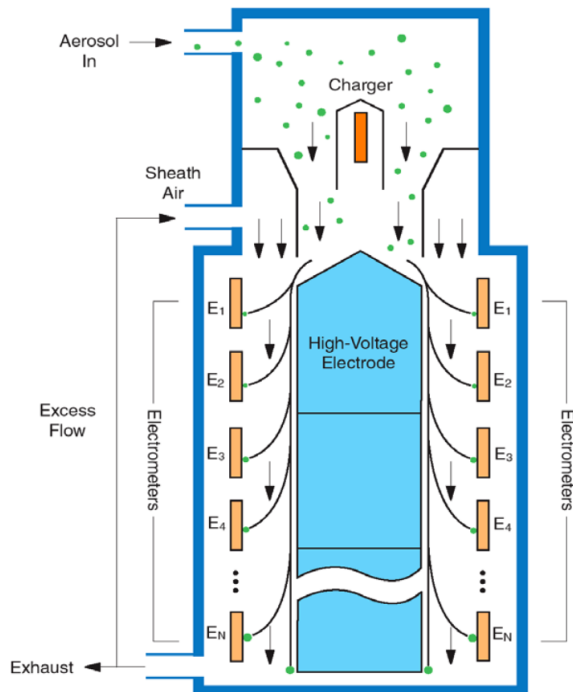


Smearing effect





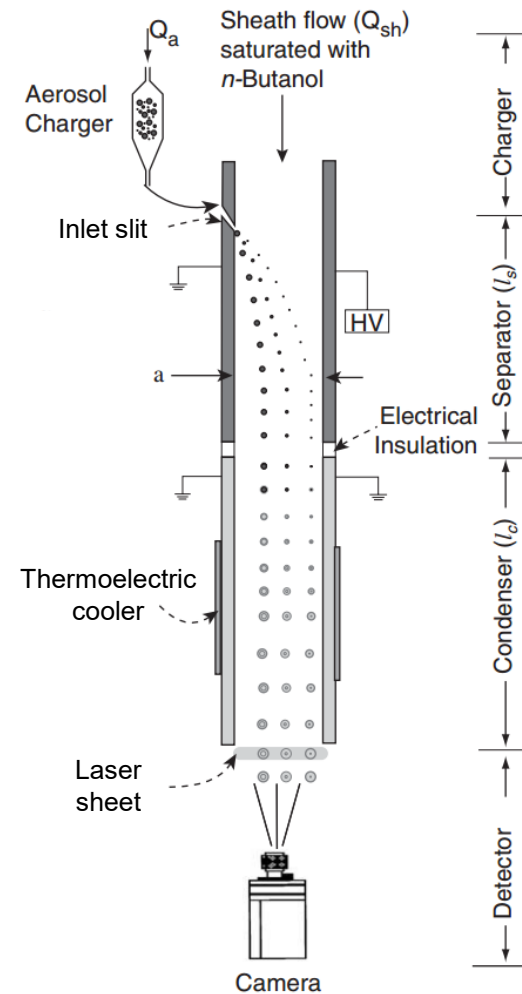
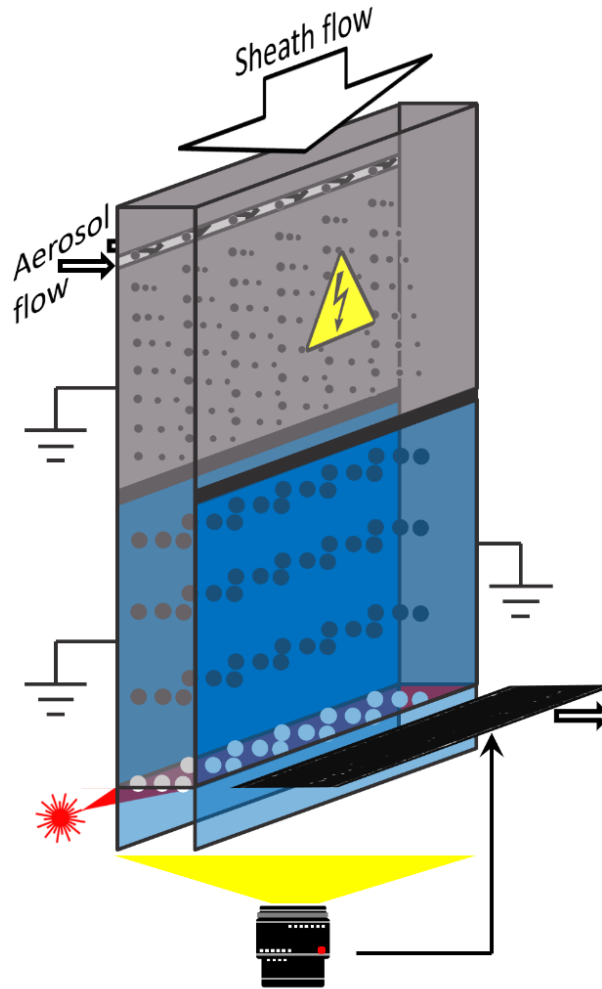
Electrical aerosol spectrometer



EEPS 3090, TSI Inc

- Engine Exhaust Particle Sizer (EEPS), Fast Mobility Particle Sizer (FMPS) by TSI, and Differential Mobility Spectrometer (DMS500, Cambustion)
- Both based on Electrical Aerosol Spectrometer (EAS) developed by Tammet et al. (2002)
- Real-time measurement of aerosol size distribution
- Limited to aerosol with high concentration (i.e., source level measurements)

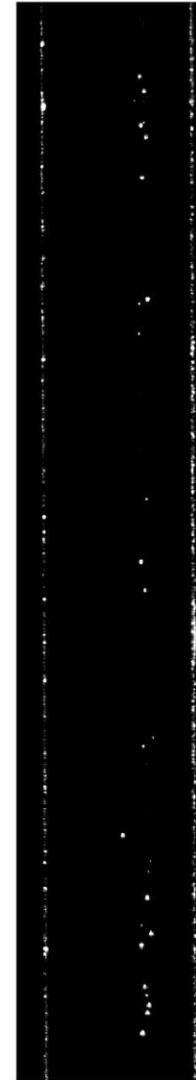
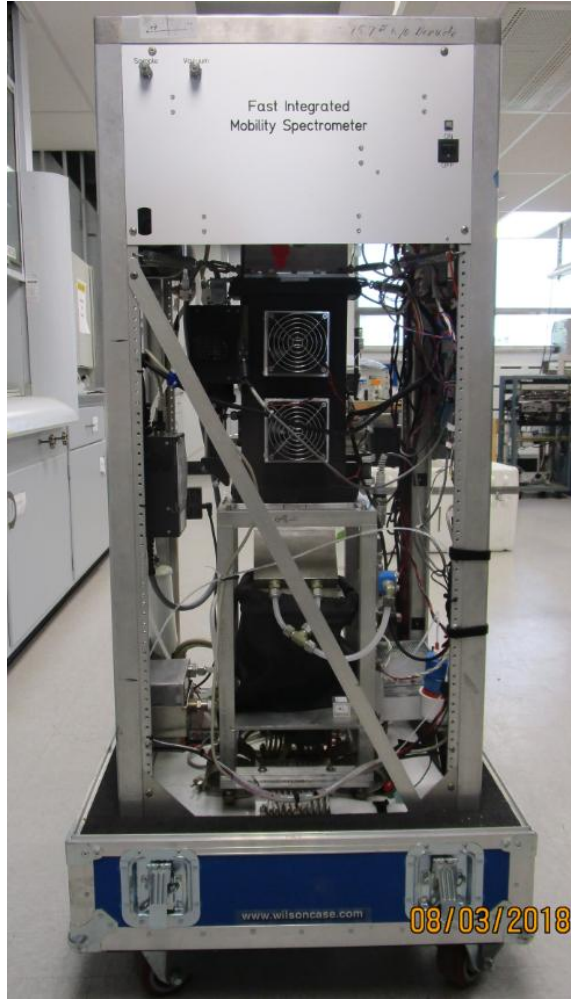
Fast integrated mobility spectrometer (FIMS):



Kulkarni, Wang (2006). J. Aerosol Sci. **37**(10): 1303-1325.

Kulkarni, Wang (2006). J. Aerosol Sci. **37**(10): 1326-1339.

Fast integrated mobility spectrometer (FIMS):



100 nm particle

Kulkarni, Wang (2006). *J. Aerosol Sci.* **37**(10): 1303-1325.

Kulkarni, Wang (2006). J. Aerosol Sci. **37**(10): 1326-1339.

Aerodynamic sizer

Aerodynamic Particle Sizer (APC)

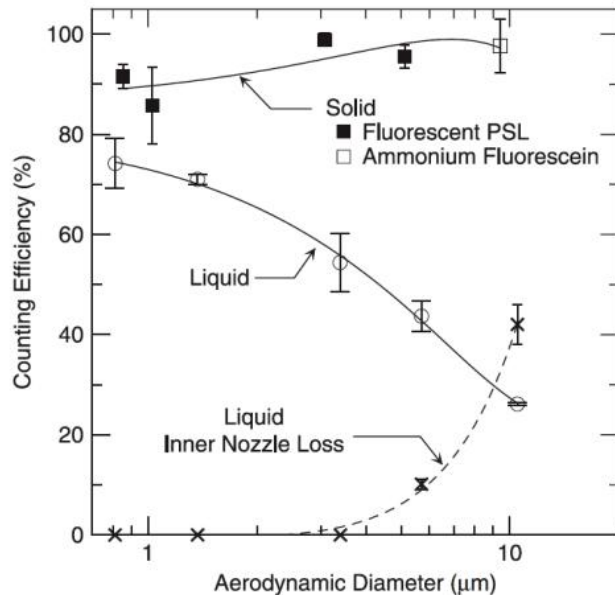
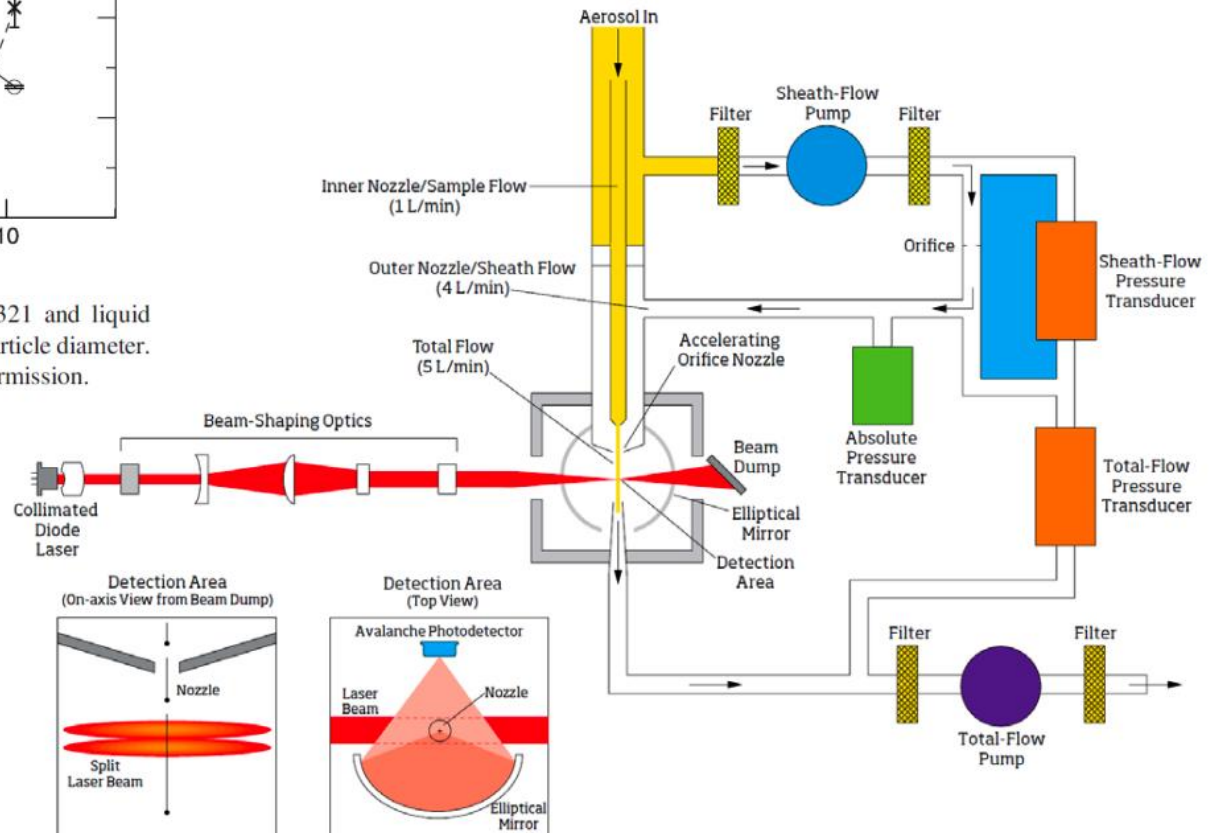


Figure 14-12 Counting efficiency of the APS 3321 and liquid losses within the inner nozzle versus aerodynamic particle diameter. Reprinted from Volckens and Peters (2005) with permission.



Issues with sizing droplets

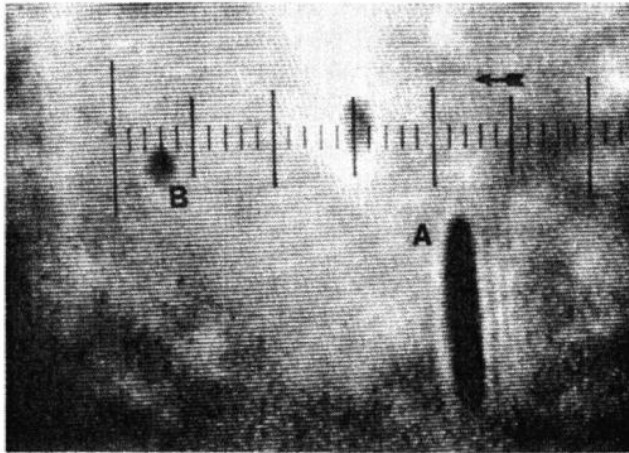


Figure 14-15 Picture of droplets in the high velocity air jet just beyond the APS nozzle taken with a high-speed laser-imaging system showing a droplet flattened by the drag force. The nozzle tip is about 200 μm to the right of droplet A with the air and the droplets moving to the left. The scale markers are approximately 5 μm apart. The larger droplet A has an extreme 10-by-60 μm spheroidal shape while the smaller droplet B is about 8 by 10 μm and is only slightly flattened.

$$\Delta = -\frac{(2.723 \times 10^{-4})d^2}{\eta^{0.6486}\gamma^{0.3864}} \quad (\text{Eq. 14-1})$$

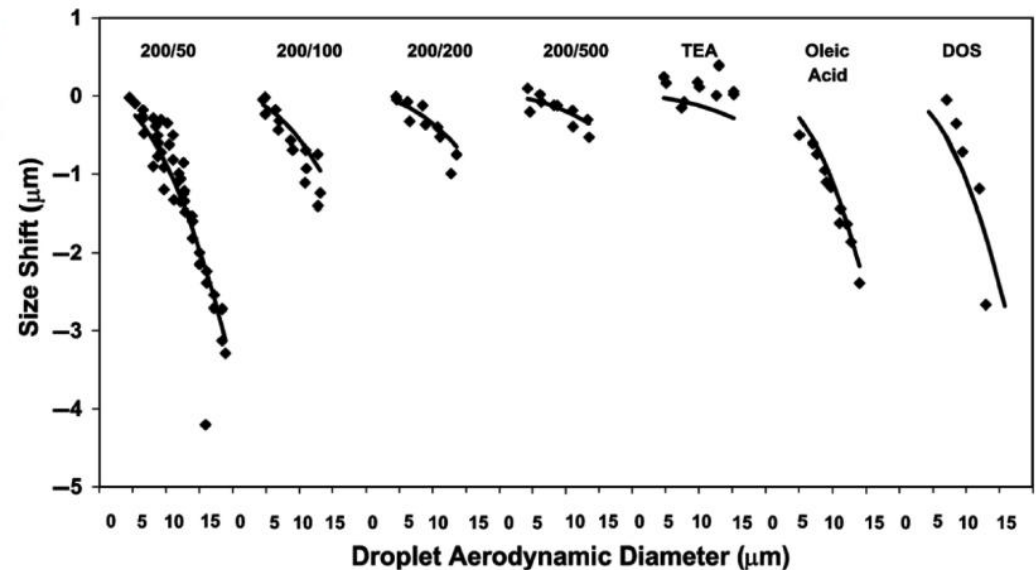
TABLE 14-2 Properties of Liquids Used in the Study of Droplet Size Shifts^a

Oil Type	Surface Tension (N/m)	Viscosity (Pa · s)
200/50 ^b	0.0208	0.048
200/100 ^b	0.0209	0.0964
200/200 ^b	0.021	0.1934
200/500 ^b	0.0211	0.4845
Triethanolamine (TEA)	0.0489	0.59
Oleic acid	0.032	0.0256
Di-octyl sebacate (DOS) ^c	0.0322	0.027

^aSee Figure 14-16.

^bDesignation for Dow Corning polydimethylsiloxane oils.

^cAlso called (bis)-2-ethyl hexyl sebacate.



Aerodynamic aerosol classifier

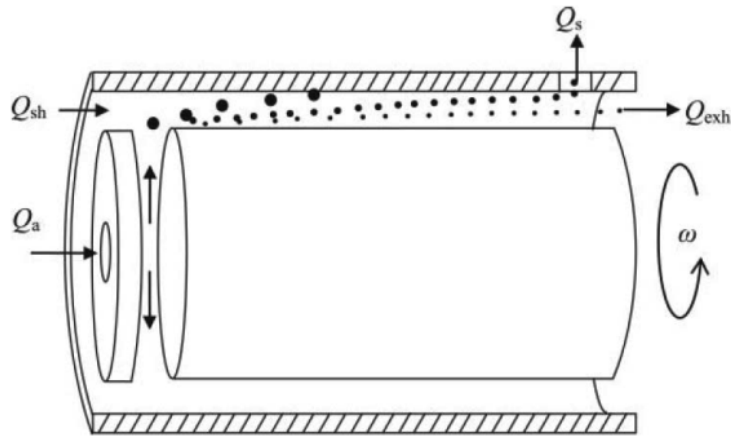


FIG. 1. Schematic of the AAC.

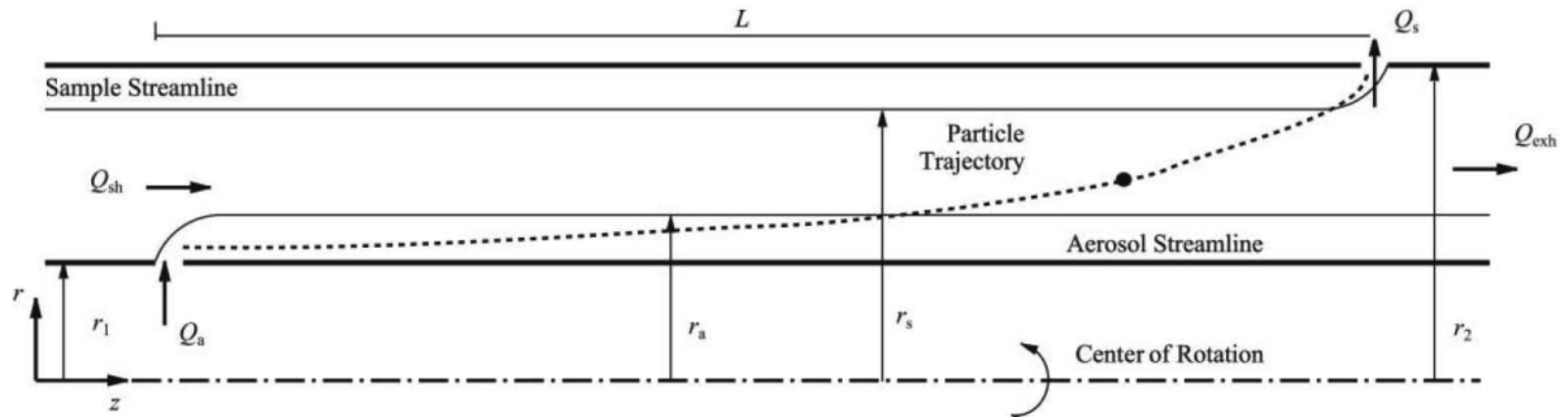
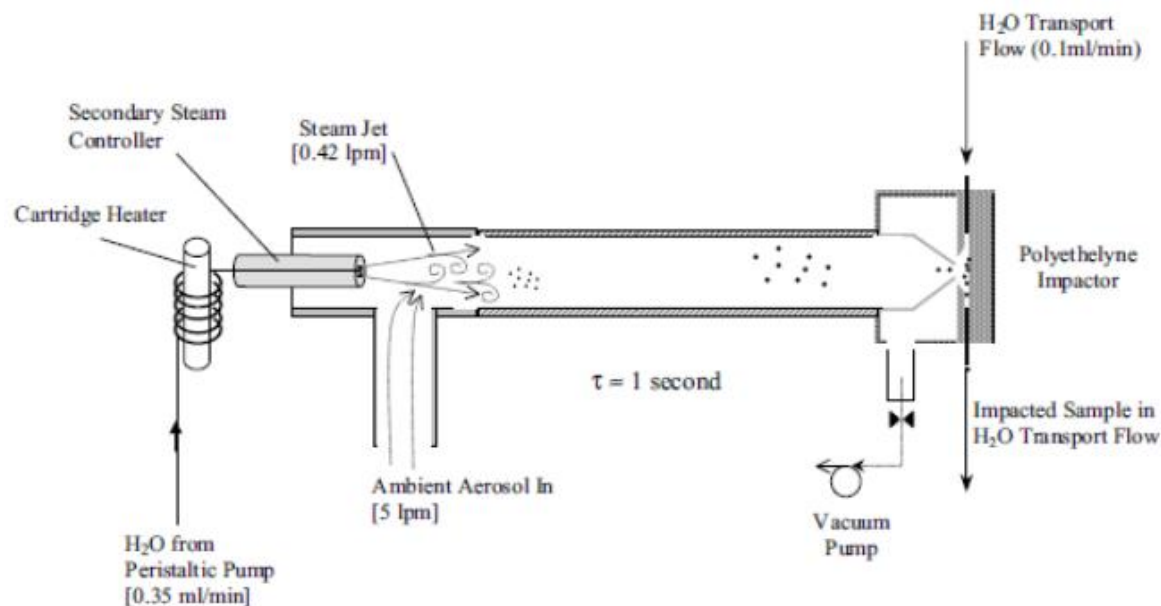


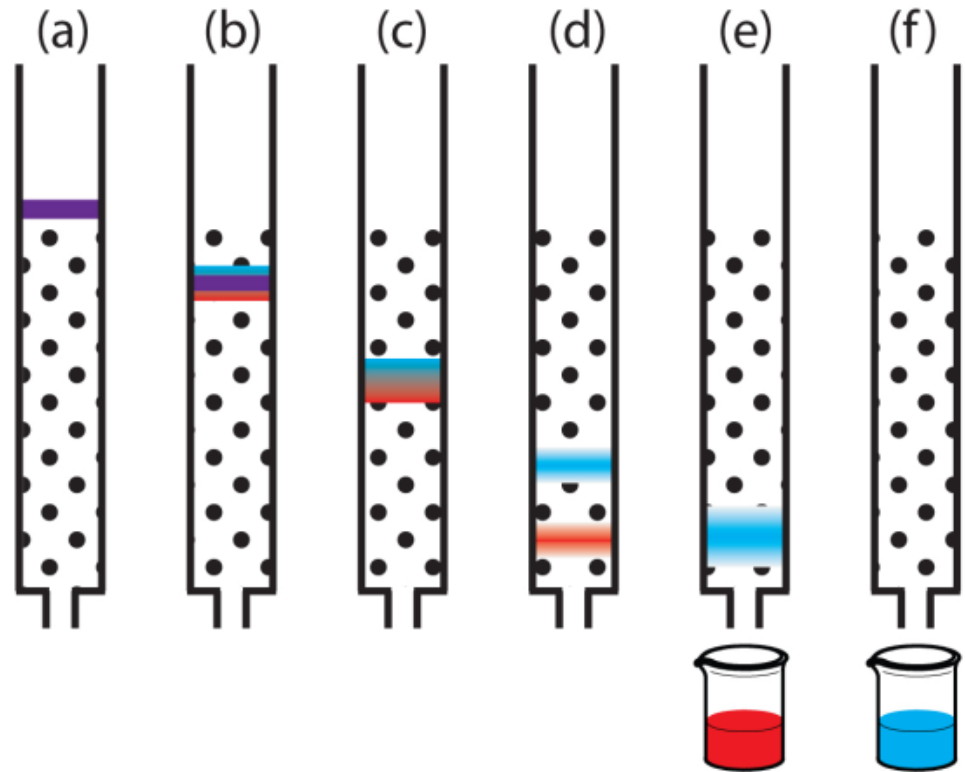
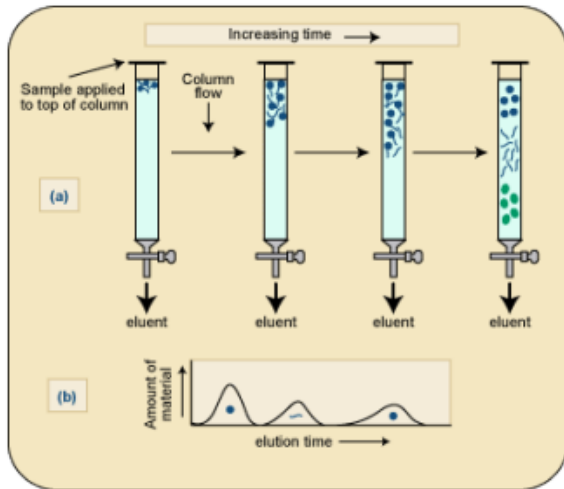
FIG. 2. Details of the particle trajectory and flows between the cylinders.

PILS

Particle-into-Liquid Sampler (PILS)

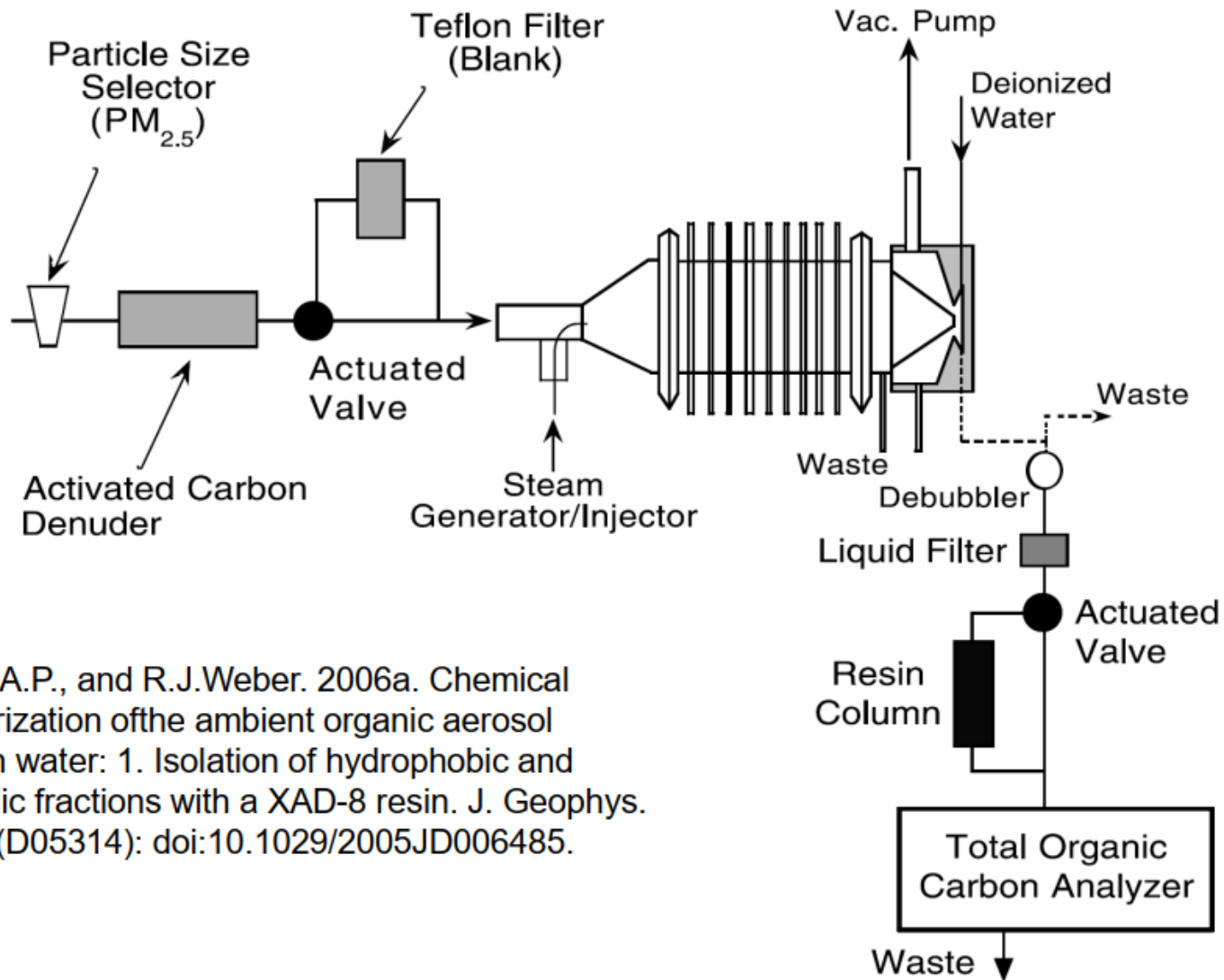


Ion Chromatography



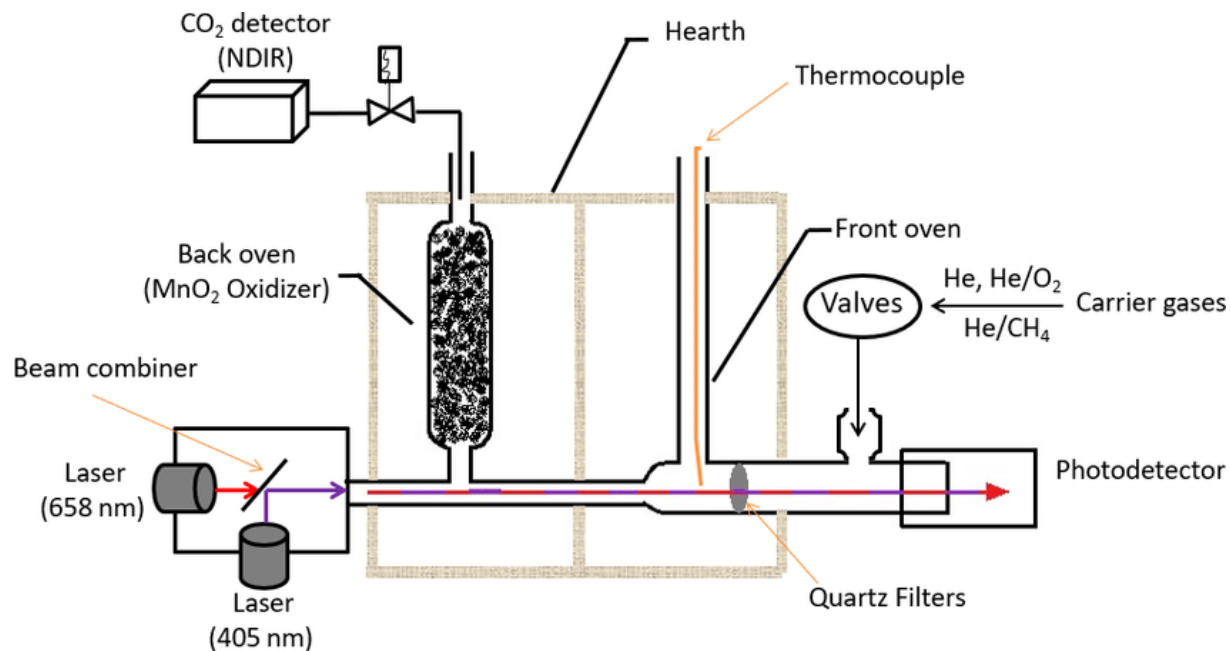
Sulfate, nitrate, ammonia, organics
Several metal ions

Water soluble organic carbon

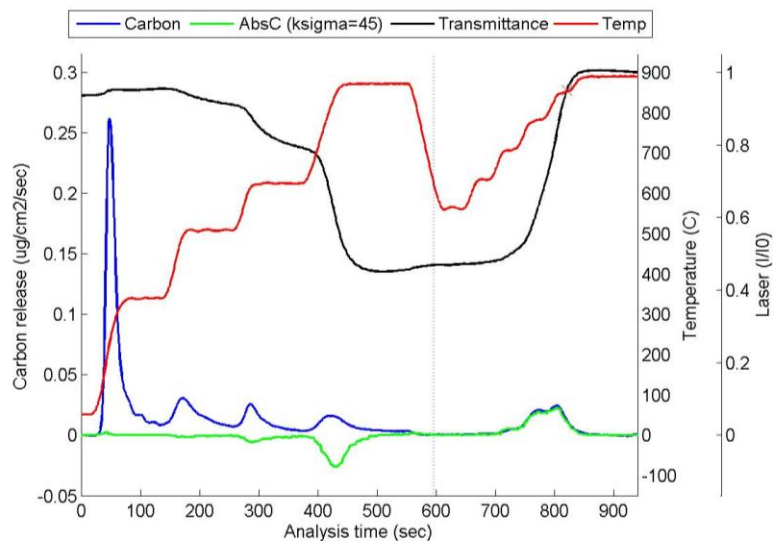


Sullivan, A.P., and R.J.Weber. 2006a. Chemical characterization of the ambient organic aerosol soluble in water: 1. Isolation of hydrophobic and hydrophilic fractions with a XAD-8 resin. J. Geophys. Res. 111(D05314): doi:10.1029/2005JD006485.

OC/EC analyzer



He: obtain OC
O₂: obtain EC



Elemental analysis

TABLE 9-2 Detection Limits for Key Atmospheric Tracers and Potentially Toxic Trace Elements in PM Collected on Filters (in ng/cm² of filter^a) as Determined by INAA, PIXE, XRF, and ICP-MS

Species	Method			
	INAA ^a	PIXE ^a	XRF ^b	ICP-MS ^c
Al	2.6	9.1	18	–
As	0.04	0.24	0.8	0.009
Ba	2.1	–		0.07
Br	62	3.2	0.6	–
Ca	–	3.6	1.5	1.2
Cd	–	–	22	0.02
Cr	6.4	4.0	3	0.08
Cu	2.4	0.39	0.7	0.03
Fe	53	2.2	0.7	11
Hg	–	–	1.5	–
Mg	53	36	3.2	0.1
Mn	0.06	0.48	0.8	0.08
Nd	–	–		–
Ni	2.5	0.39	0.6	0.06
Pb	–	–	1.5	0.007
S	–	–	2.6	–
Sb	–	–	31	0.021
Se	0.48	0.26	0.7	0.12
Sr	9.6	0.48	1.1	0.008
Ti	10	1.2	16	0.11
V	0.03	0.76	5.3	0.04
Zn	2.6	0.29	1	0.08

^aCalculated from Salma et al. (1997).

^bData from USEPA (1999a).

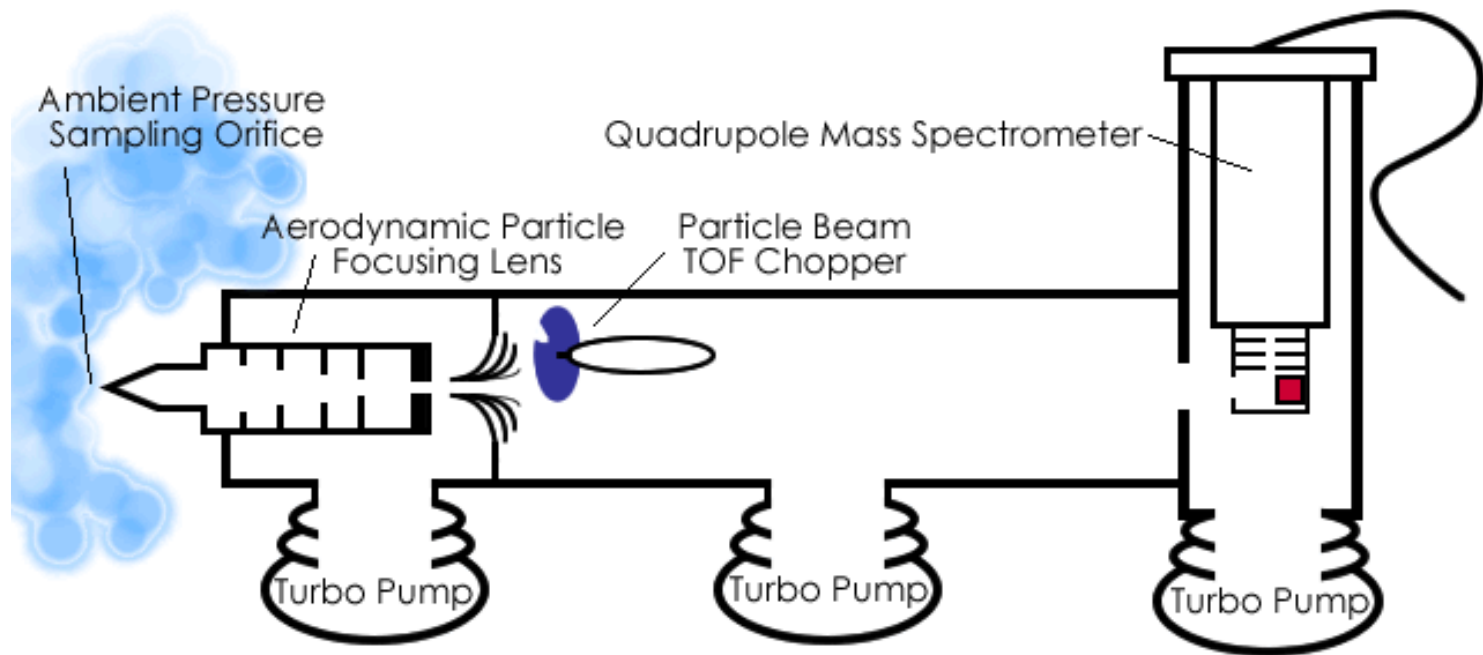
^cData from Pekney and Davidson (2005).

- X-ray fluorescence (XRF)
- Inductively coupled plasma-mass spectrometry (ICP-MS)

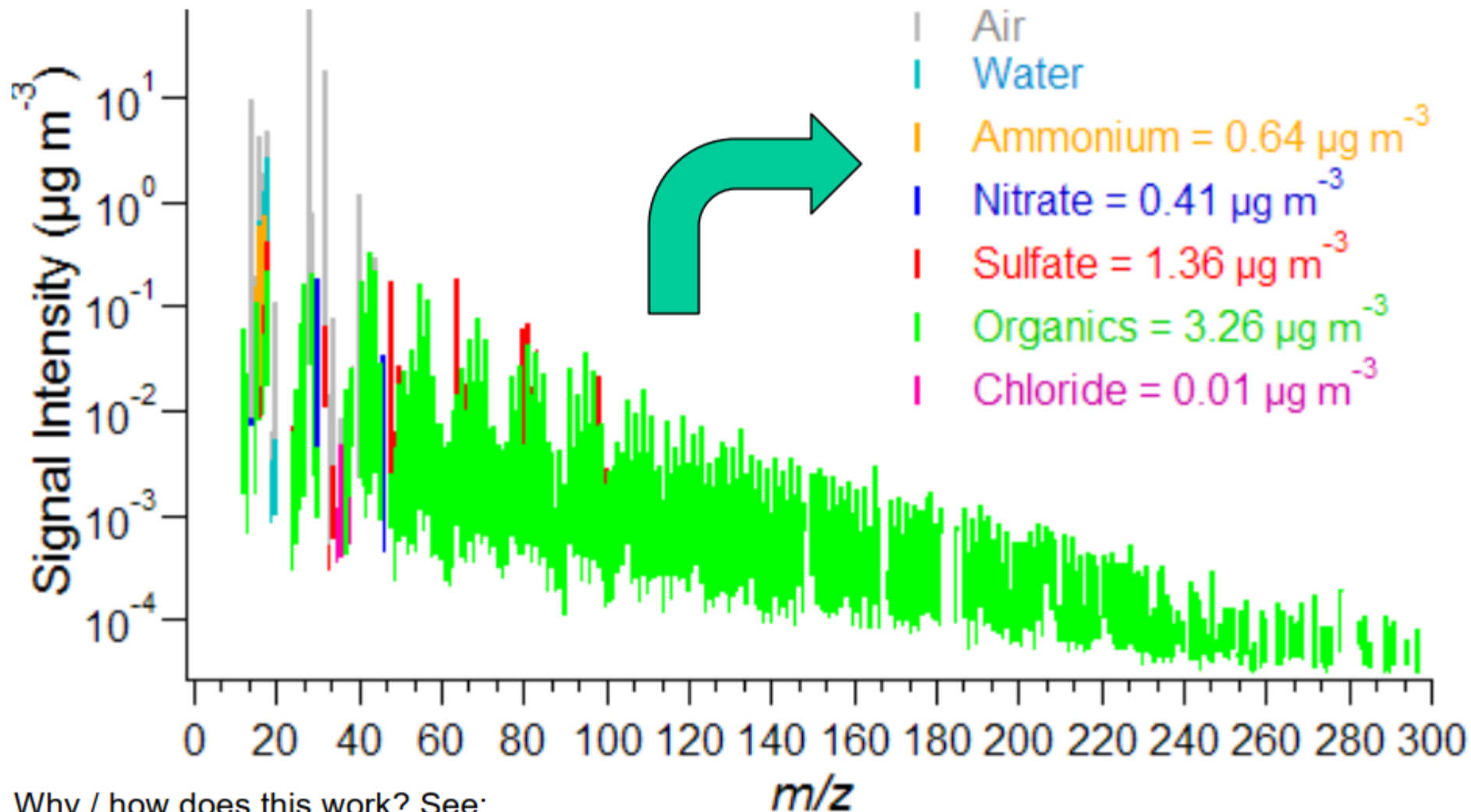
AMS

How AMS works

- Example: Aerodyne AMS



Ambient mass spectrum



Why / how does this work? See:

- Jimenez et al. *J. Geophys. Res.*, 108, 8425, 2003.
- Allan et al., *J. Aerosol Sci.*, 35, 909, 2004.
- Canagaratna et al., *Mass. Spec. Rev.*, 26, 185-222, 2007

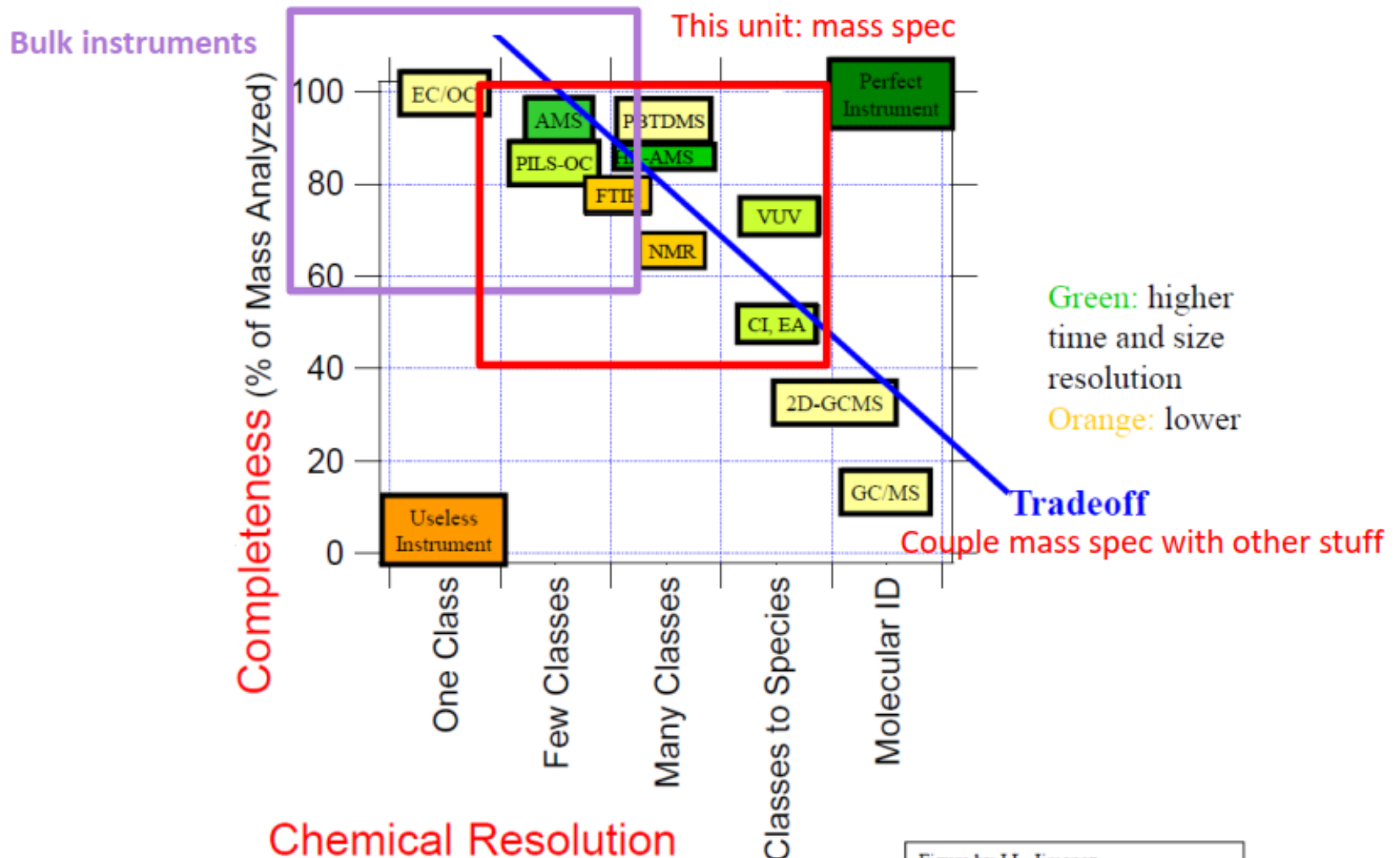
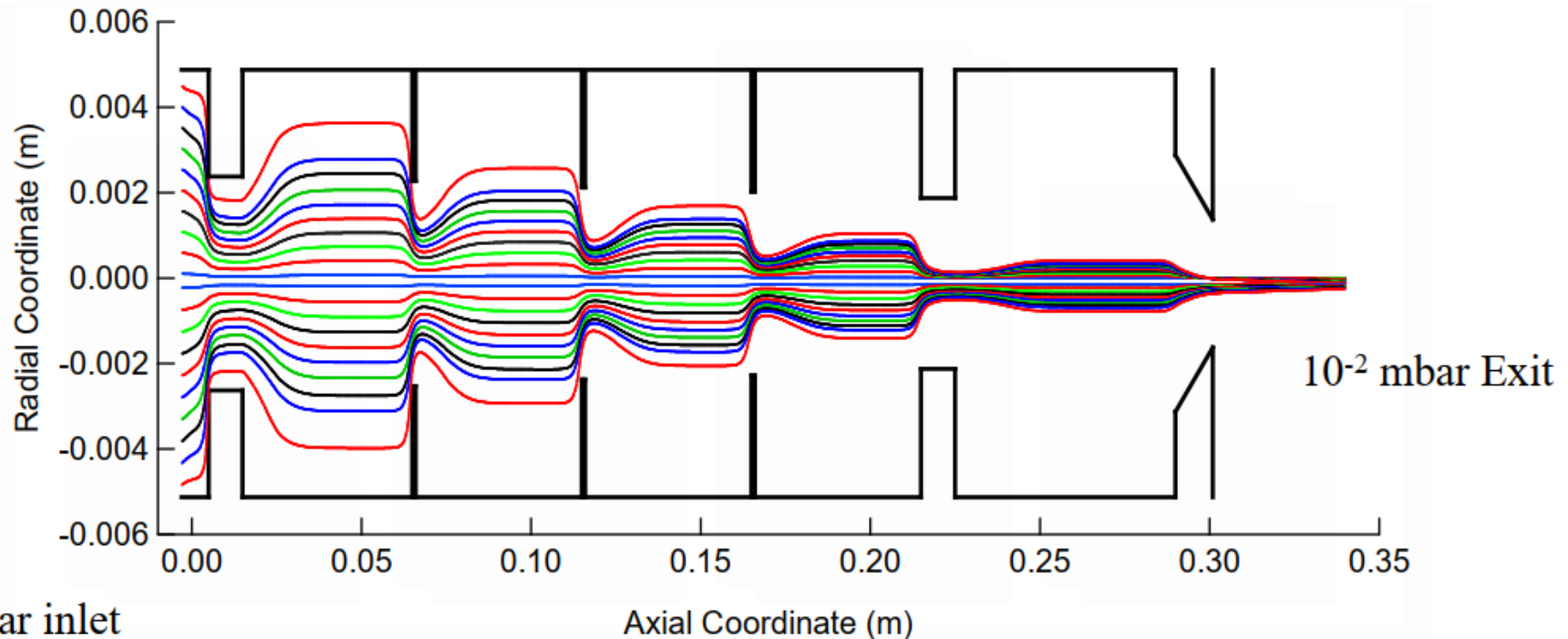


Figure by J.L. Jimenez
Published in Hallquist et al., ACP 2009

<https://www.youtube.com/watch?v=kmZHwCBJHO8>

Aerodynamic Lense



**Calculated Particle Trajectories, 100 nm (Fluent ver 4.47)
Diameter Unit Density Spheres**

Original lens design:

- Liu, P., Ziemann, P. L., Kittelson, D. B., and McMurry, P. H. *Aerosol Sci. Technol.* 22:293–313, 1995.
- Liu, P., Ziemann, P. L., Kittelson, D. B., and McMurry, P. H. *Aerosol Sci. Technol.* 22:314–324, 1995.

CFD Simulations:

- Zhang, X., Smith, K.A., Worsnop, D.R., Jimenez, J.L., Jayne, J.T., and Kolb, C.E. *Aerosol Sci. Technol.*, 36: 617, 2002.
- Zhang, X., Smith, K.A., Worsnop, Jimenez, J.L., Jayne, J.T., D.R., Kolb, C.E., Morris, J., Davidovits, P., *Aerosol Sci. Technol.*, 38: 619, 2004.

Beam width w/ aerodynamic Lense

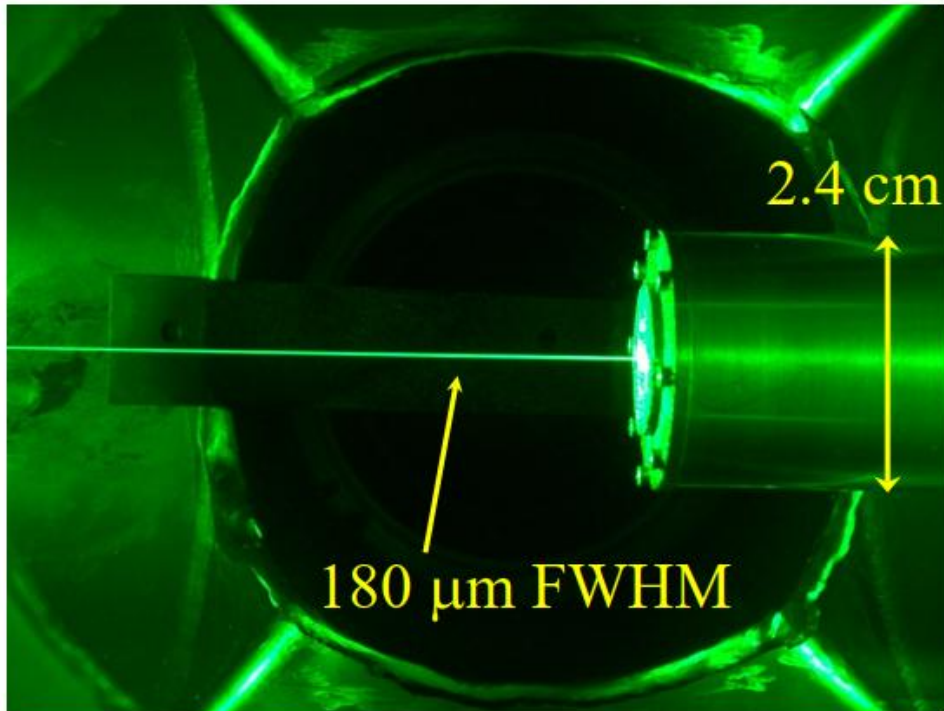
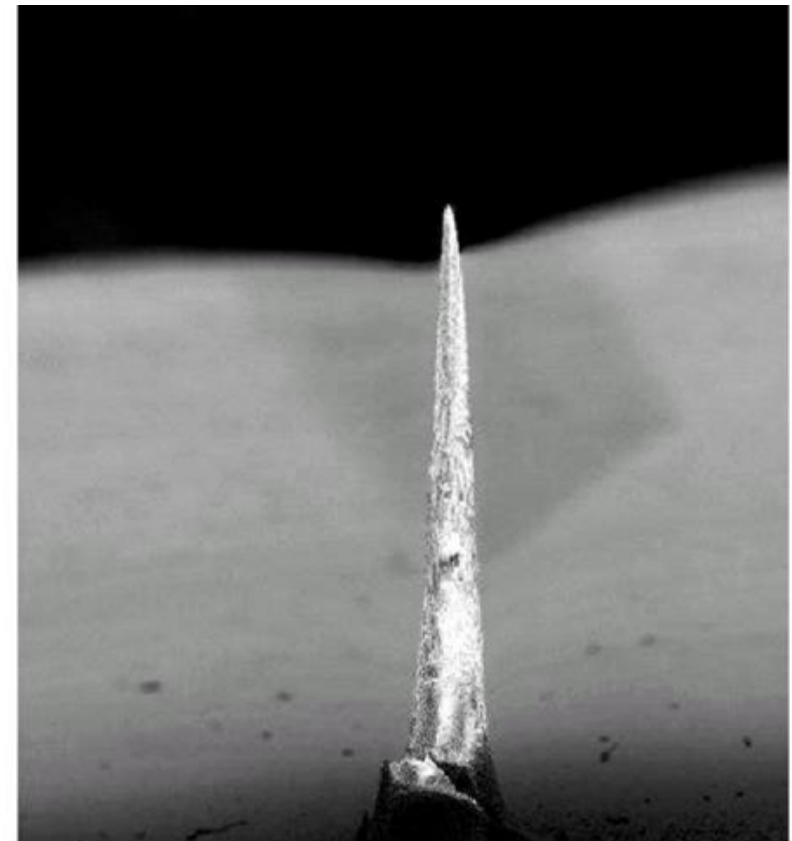


Figure courtesy of Guiseppe Petrucci, Univ. Vermont
<http://www.uvm.edu/~gpetrucc>

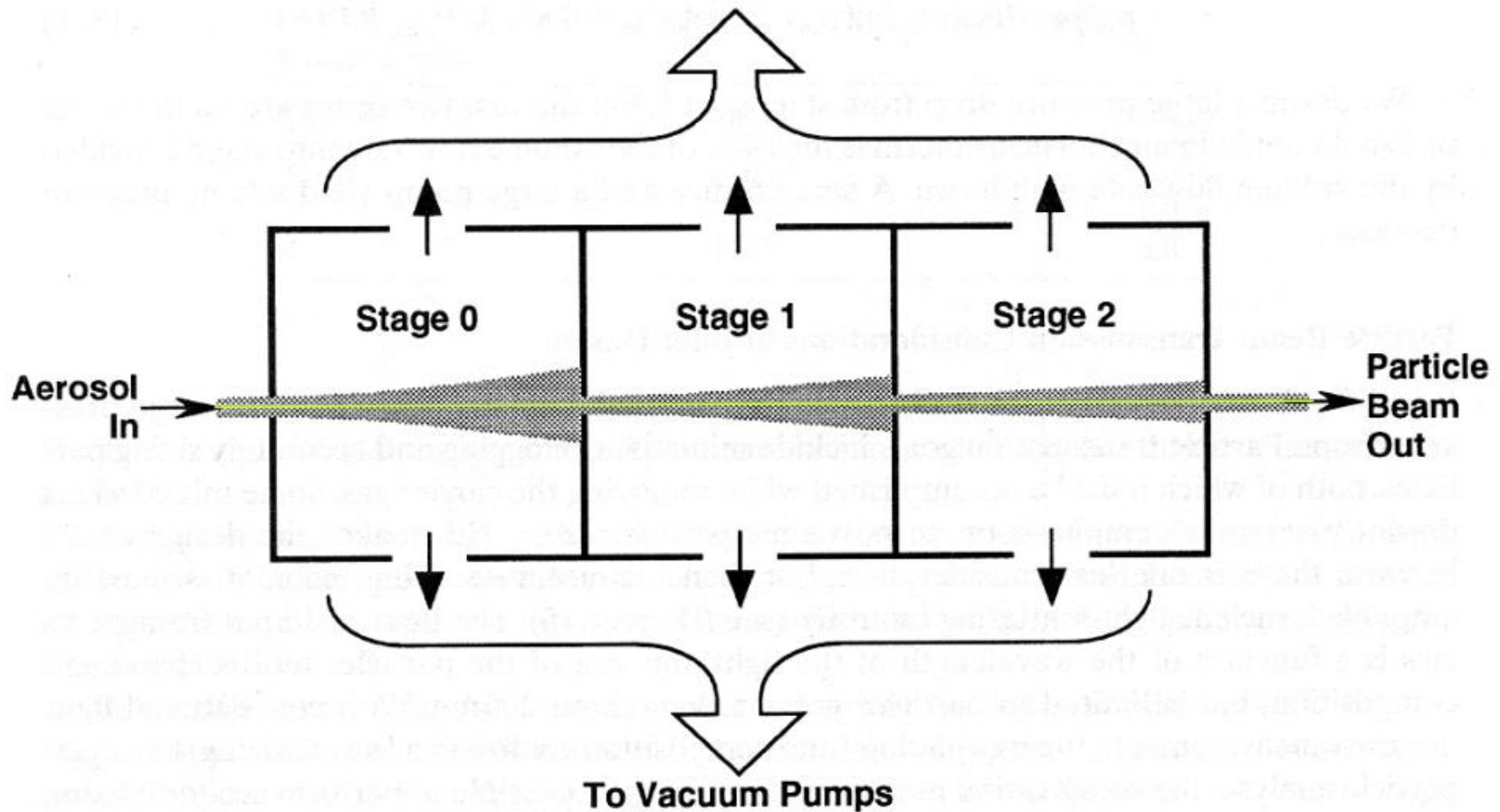


400μm 65X

Fig. 6. SEM image of a silicon carbide 'tower' deposited by a focused nanoparticle beam [18].

Heberlein, J., O. Postel, S. Girshick, P. McMurry, W. Gerberich, D. Iordanoglou, F.D. Fonzo, D. Neumann, A. Gidwani, M. Fan, and N. Tymiak, Thermal plasma deposition of nanophase hard coatings. *Surface and Coatings Technology*, 2001. **142-144**: p. 265-271.

Differential pumping of gas



- E.g. in AMS: particles concentrated by 10^7 with respect to gas

Wexler, A. S., and Johnston, M. V. (2001). "Real-time single-particle analysis." *Aerosol Measurement: Principles, Techniques, and Applications*, P. A. Baron and K. Willeke, eds., Wiley-Interscience, New York, 365-386.